Phase 1

```
chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

Phase 2

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes commit -m "Hello"
Committed: 6d05de07d3d4... Hello
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ find .pes/objects -type f
.pes/objects/31/1a8087c59ba9f4f174e9278c44df319489fc86c93ddf67e688e60da23d8baa
.pes/objects/6d/05de07d3d4150f153ea324cb91c179b92b02f363512e0f8018438c983720be
.pes/objects/7c/5c8610459154bdde4984be72c48fb5d9c1c4ac793a6b5976fe38fd1b0b1284
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ xxd .pes/objects/31/1a8087c59ba9f4f174e9278c44df319489fc86c93ddf67e688
e60da23d8baa | head -20
00000000: 7472 6565 2034 3900 3130 3036 3434 2068  tree 49.100644 h
00000010: 656c 6c6f 2e74 7874 007c 5c86 1045 9154  ello.txt.|\..E.T
00000020: bdde 4984 be72 c48f b5d9 c1c4 ac79 3a6b  ..I..r.......y:k
00000030: 5976 fe38 fd1b 0b12 84                   Yv.8.....
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

Phase 3

```
chandak06@Ubuntu: ~/Desktop/sdn-project        chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ echo "hello" > file1.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ echo "world" > file2.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes add file1.txt file2.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  README.md
  untracked:  pes.h
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  index.h
  untracked:  index.c
  untracked:  tree.c
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  report.md
  untracked:  Makefile
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  tree.h
  untracked:  object.c
  untracked:  commit.h
```

```
chandak06@Ubuntu: ~/Desktop/sdn-project        chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  README.md
  untracked:  pes.h
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  index.h
  untracked:  index.c
  untracked:  tree.c
  untracked:  test_objects.c
  untracked:  commit.c
  untracked:  report.md
  untracked:  Makefile
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  tree.h
  untracked:  object.c
  untracked:  commit.h

chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ cat ./pes/index
cat: ./pes/index: Not a directory
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ cat .pes/index

100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776663890 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776663897 6 file2.txt
100644 7c5c8610459154bdde4984be72c48fb5d9c1c4ac793a6b5976fe38fd1b0b1284 1776662002 12 hello.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

Phase 4

chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs

chandak06@Ubuntu: ~/Desktop/sdn-project | chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ echo "Hello" > hello.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes add hello.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes commit -m "Initial commit"
Committed: 687f3a13048e... Initial commit
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ echo "World" >> hello.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes add hello.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes commit -m "Add world"
Committed: 50113705eefe... Add world
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
echo "Goodbye" > bye.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes add bye.txt
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes commit -m "Add farewell"
Committed: d93985f71eab... Add farewell
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ ./pes log
commit d93985f71eabebb335eadcba3b76b6ede1cab3d0d0d6f63155c12a1ffa79b8e2
Author: PES User <pes@localhost>
Date:   1776664237

    Add farewell

commit 50113705eefe871ee576eac7d256365a29889c3892c22685f6d1763718882772
Author: PES User <pes@localhost>
Date:   1776664213

    Add worldell

commit 687f3a13048e87e15e97afc956044b851d7b08593d076ebc1799b2fcffa9c246
Author: PES User <pes@localhost>
Date:   1776664190
```

chandak06@Ubuntu: ~/Desktop/sdn-project | chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs

```
chandak06@U[illegible]es-vcs$ ./pes log
commit d93[illegible]ab3d0d0d6f63155c12a1ffa79b8e2
Author: PES User <pes@localhost>
Date:   1776664237

    Add farewell

commit 50113705eefe871ee576eac7d256365a29889c3892c22685f6d1763718882772
Author: PES User <pes@localhost>
Date:   1776664213

    Add worldell

commit 687f3a13048e87e15e97afc956044b851d7b08593d076ebc1799b2fcffa9c246
Author: PES User <pes@localhost>
Date:   1776664190

    Initial commit

commit 6d05de07d3d4150f153ea324cb91c179b92b02f363512e0f8018438c983720be
Author: PES User <pes@localhost>
Date:   1776662077

    Hello207a
```

```
chandak06@Ubuntu: ~/Desktop/sdn-project        chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs
chandak06@Ubuntu: ~/Desktop/sdn-project
chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs$ ./pes log
commit d93985f71eabebb335eadcba3b76b6ede1cab3d0d0d6f63155c12a1ffa79b8e2
Author: PES User <pes@localhost>
Date:   1776664237

    Add farewell

commit 50113705eefe871ee576eac7d256365a29889c3892c22685f6d1763718882772
Author: PES User <pes@localhost>
Date:   1776664213

    Add worldell

commit 687f3a13048e87e15e97afc956044b851d7b08593d076ebc1799b2fcffa9c246
Author: PES User <pes@localhost>
Date:   1776664190

    Initial commit

commit 6d05de07d3d4150f153ea324cb91c179b92b02f363512e0f8018438c983720be
Author: PES User <pes@localhost>
Date:   1776662077

    Hello207a
```

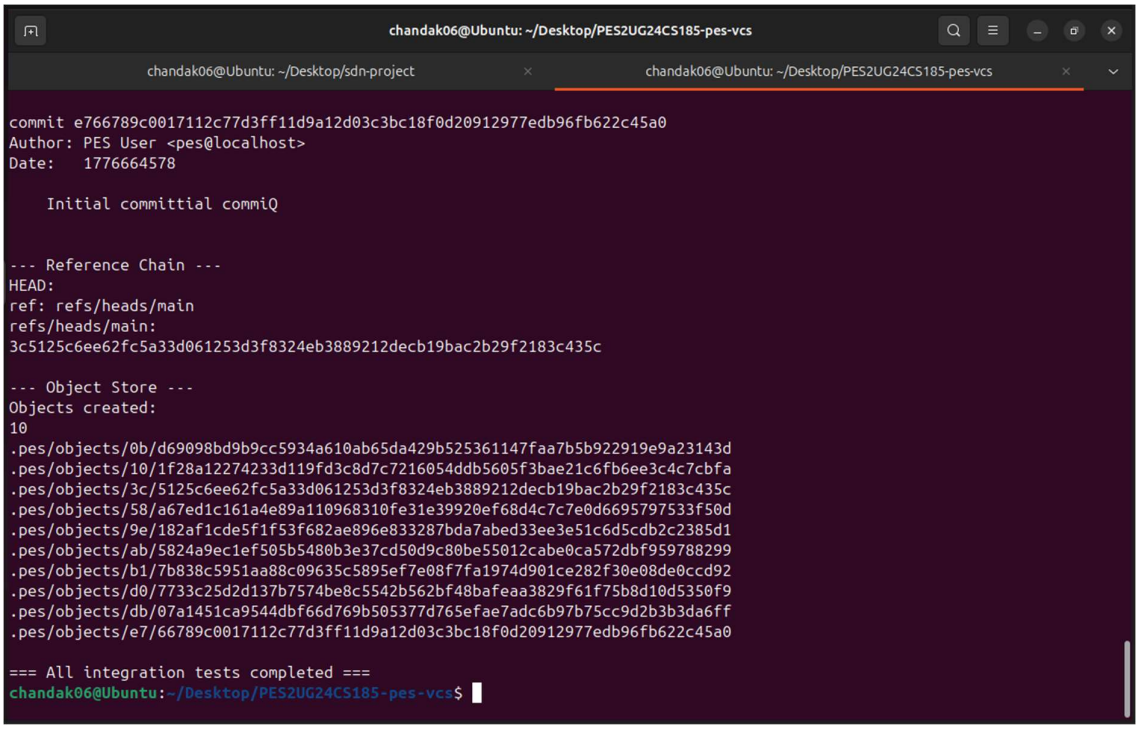
```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/31/1a8087c59ba9f4f174e9278c44df319489fc86c93ddf67e688e60da23d8baa
.pes/objects/50/113705eefe871ee576eac7d256365a29889c3892c22685f6d1763718882772
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/68/7f3a13048e87e15e97afc956044b851d7b08593d076ebc1799b2fcffa9c246
.pes/objects/6d/05de07d3d4150f153ea324cb91c179b92b02f363512e0f8018438c983720be
.pes/objects/7c/5c8610459154bdde4984be72c48fb5d9c1c4ac793a6b5976fe38fd1b0b1284
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/d9/3985f71eabebb335eadcba3b76b6ede1cab3d0d0d6f63155c12a1ffa79b8e2
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ cat .pes/refs/heads/main
d93985f71eabebb335eadcba3b76b6ede1cab3d0d0d6f63155c12a1ffa79b8e2
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

MAKE TEST INTEGRATION

```
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: e766789c0017... Initial commit

Log after first commit:
commit e766789c0017112c77d3ff11d9a12d03c3bc18f0d20912977edb96fb622c45a0
Author: PES User <pes@localhost>
```

```
chandak06@Ubuntu: ~/Desktop/sdn-project    chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs
Author: PES User <pes@localhost>
Date:   1776664578

    Initial commit


--- Second Commit ---
Committed: 9e182af1cde5... Update file.txt

--- Third Commit ---
Committed: 3c5125c6ee62... Add farewell

--- Full History ---
commit 3c5125c6ee62fc5a33d061253d3f8324eb3889212decb19bac2b29f2183c435c
Author: PES User <pes@localhost>
Date:   1776664578

    Add farewell

commit 9e182af1cde5f1f53f682ae896e833287bda7abed33ee3e51c6d5cdb2c2385d1
Author: PES User <pes@localhost>
Date:   1776664578

    Update file.txt

commit e766789c0017112c77d3ff11d9a12d03c3bc18f0d20912977edb96fb622c45a0
Author: PES User <pes@localhost>
Date:   1776664578

    Initial committial commiQ
```



```
chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs
chandak06@Ubuntu: ~/Desktop/sdn-project    chandak06@Ubuntu: ~/Desktop/PES2UG24CS185-pes-vcs

commit e766789c0017112c77d3ff11d9a12d03c3bc18f0d20912977edb96fb622c45a0
Author: PES User <pes@localhost>
Date:   1776664578

    Initial committial commiQ


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
3c5125c6ee62fc5a33d061253d3f8324eb3889212decb19bac2b29f2183c435c

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3c/5125c6ee62fc5a33d061253d3f8324eb3889212decb19bac2b29f2183c435c
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/9e/182af1cde5f1f53f682ae896e833287bda7abed33ee3e51c6d5cdb2c2385d1
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e7/66789c0017112c77d3ff11d9a12d03c3bc18f0d20912977edb96fb622c45a0

=== All integration tests completed ===
chandak06@Ubuntu:~/Desktop/PES2UG24CS185-pes-vcs$
```

Q5.1: In PES-VCS, pes checkout <branch> would first read the commit hash stored in .pes/refs/heads/<branch> and update .pes/HEAD to point to that branch. Then it must load the target commit's tree, update the working directory files to match that snapshot, remove files no longer present, and recreate the index accordingly. This operation is complex because it changes HEAD, index, and actual files together. It must also avoid overwriting user changes or deleting important data accidentally.

Q5.2: To detect a dirty working directory, compare each tracked file in the index with the current working file. If size or modification time differs, recompute the file's hash and compare it with the hash stored in .pes/index. If they differ, the file has uncommitted changes. Then compare that file with the target branch's version from the object store. If checkout would overwrite those edits, the command must refuse and warn the user.

Q5.3: In detached HEAD state, .pes/HEAD stores a commit hash directly instead of a branch reference. New commits can still be created, but no branch name moves forward to track them. If the user switches branches later, those commits may become unreachable. Recovery is possible by creating a new branch pointing to those commits or using commit hashes from logs/reflog-like history. Detached HEAD is useful for temporary experiments.

Q6.1: Garbage collection would start from all branch heads in .pes/refs/heads/ and mark every reachable commit, tree, and blob by recursively following parent links and tree references. After marking, any object file not visited is unreachable and can be

deleted. A hash set is ideal for storing visited hashes efficiently. For 100,000 commits and 50 branches, roughly all reachable commits plus their trees and blobs must be scanned, so several hundred thousand objects may be visited.

Q6.2: Running garbage collection during a commit is dangerous because GC may delete objects created by the commit before the branch reference is updated. For example, a tree object may be written, but before the new commit points to it, GC sees no reference and removes it. Then the commit finishes with missing objects, corrupting history. Real Git avoids this using locks, temporary files, and only deleting objects older than a safety grace period.